Bob Ma

Senior II Machine Learning Engineer · ML Pipelines & MLOps · Data Platforms · Production ML

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§ 01 SUMMARY

Auckland-based machine learning engineer with a PhD from AUT, 6+ years shipping production ML and 18+ years in software and data engineering. Specialises in the *data pipelines and automation* behind production ML: ETL pipelines on AWS (SageMaker, Lambda, Glue, S3) and Snowflake, Spark processing orchestrated with Airflow, automated training, evaluation and deployment, and Concourse CI/CD that provisions infrastructure and ships models behind automated validation. Applies ML from forecasting and behaviour modelling to multimodal LLMs and VLMs; at Resideo, automated VLM-based business assessment for small and medium-sized business (SMB) customers, from front-of-store data through to POS, payment and alerting systems. Delivered across smart-home devices (Resideo / Honeywell Home), commercial transport (EROAD), retail (Woolworths NZ, StayinFront) and construction (BuildingEstimates); inventor on three patents and reviewer for AAAI, ACM TOMM and IEEE journals.

§ 02 RELEVANT STRENGTHS

ML PIPELINES & MLOPS
ETL pipelines feeding continuous training, evaluation and deployment; Concourse CI/CD automating infrastructure provisioning and model deployment; automated validation as a release gate.

DATA PLATFORMS
AWS SageMaker, Lambda, Glue and S3; Spark processing with Airflow orchestration; Snowflake and SQL; data-warehouse and SAP / ERP integration at Woolworths NZ.

PRODUCTION ENGINEERING & DEVOPS
Python and C++ services (Flask, REST); Docker and Kubernetes; unit, stress and automated CI/CD testing; Git; Agile / Scrum delivery.

SMB BUSINESS FLOWS
Front-of-store information (shopfront photos, shelf and price-tag images, camera video, POS and inventory data) connected to POS, payment, alerting, ticketing and app systems; SMB fleet driving and NZTA inspection data; integration with company and client ERP systems.

VLMS & GENAI
VLM business assessments at Resideo (merchant verification, operating status and risk, shelf compliance, security events), configured per customer through rules and prompt templates; multimodal LLMs via REST APIs; Claude Agent SDK for code review and data processing.

FORECASTING & APPLIED ML
Demand, price and sales forecasting; driving-behaviour modelling (LSTM, k-means); compression, quantisation and distillation to meet production budgets.

§ 03 EXPERIENCE

Senior II Machine Learning Engineer

Mar 2024 — Oct 2026

Resideo Inc. (Honeywell Home) · NYSE:REZI

- *Production ML: model pipeline, VLM solutions for SMB customers and system integration*
- Designed and optimised multimodal LLM inference (LLaMA3, GPT, CLIP) and integrated model outputs with backend decision systems through REST APIs and edge–cloud pipelines on Azure.
- Owned the end-to-end model pipeline: PyTorch training and optimisation, ONNX export and conversion, experiment and model tracking in MLflow, Docker images released through GitHub Actions, and serving on Azure ML endpoints backed by a vector database.

- Applied the Claude Agent SDK to automated code review in the CI pipeline and to parts of the user-flow data processing, including building the vector-database indexes.
- Led low-level optimisation of transformer-backbone inference on edge devices: quantisation-aware deployment, mixed-precision execution, memory-access optimisation and hardware-aligned scheduling, within strict latency, memory and power budgets.
- Optimised critical kernels and adapted model operators for target chips (Ambarella CV25 / CV72 class) and their embedded inference runtimes.
- For small and medium-sized business (SMB) customers, built front-of-store information pipelines combining shopfront and signage photos, shelf and price-tag images, in-store camera video and store operating data (POS, inventory, pricing).
- Applied vision–language models (VLMs) to business assessments: merchant verification and qualification, operating-status and risk assessment, shelf-display compliance, and security-event analysis with alert summaries.
- Connected assessment results to POS and payment systems, alerting, ticketing and mobile apps; per-customer rules, categories and VLM prompt templates met each SMB customer's custom requirements.
- Built preprocessing, postprocessing, validation and automated CI/CD test pipelines (unit and stress) for production reliability.

Stack: Python · C++ · PyTorch · ONNX · VLMs · Multimodal LLMs · Claude Agent SDK · CLIP · REST APIs · **ML DevOps:** GitHub Actions · Docker · MLflow · Azure ML endpoints · Vector database

Senior Machine Learning Engineer

Mar 2022 — Mar 2024

EROAD Ltd · ASX:ERD

- *ML data platform, ETL pipelines and CI/CD for fleet-scale models*
- Developed and maintained production ETL pipelines on SageMaker, Lambda, Glue, S3, Snowflake and Docker, feeding continuous model training, evaluation and deployment.
- Built Spark processing with Airflow orchestration for fleet itinerary management.
- Used Concourse CI/CD to automate infrastructure provisioning and model deployment, keeping environments reproducible across development, testing and production.
- Processed SMB fleet customers' driving data together with NZTA vehicle-inspection data for vehicle-status and driving-behaviour models.
- Shipped real-time perception (YOLO/SSD detection, tracking, road segmentation) and collision-warning modules across large commercial vehicle fleets.

Stack: Python · C++ · PyTorch · AWS (SageMaker, Lambda, Glue, S3) · PySpark · Airflow · Snowflake · Concourse · Docker · CI/CD

Machine Learning Engineer

Jun 2021 — Feb 2022

StayinFront Ltd

- Processed front-of-store data for retail and SMB clients (shelf and price-tag images, inventory, pricing, shelf life) for sales forecasting, and integrated the results with StayinFront's own ERP and with clients' ERP systems.
- Developed the Python backend and Flask services connecting to existing AWS pipelines (S3, EC2, DynamoDB).

Stack: Python · TensorFlow / Keras · PySpark · DynamoDB · Flask · AWS

Machine Learning Engineer (Contract)

Sep 2020 — Apr 2021

BuildingEstimates.com

- Designed the framework architecture; Python/Flask services on GCP with Kubernetes processing images from Cloud Storage.
- Built plan-recognition and information-extraction models for architectural drawings (elevation, floor and roof plans) for quantity surveying.

Stack: Python · pandas · NumPy · Flask · GCP · Kubernetes

Data Engineer

Sep 2017 — Sep 2019

Woolworths NZ · Woolworths Group ASX:WOW

- Designed and developed data pipelines that deliver accurate data to downstream applications.
- Built RESTful API integrations with third-party ERP platforms (SAP portal) for data-warehouse, logistics and payment workflows.

Stack: C# · Python · Docker · Oracle · SAP · MS SQL Server

Full Stack Developer · New Image International — Jun 2014 – May 2017

E-commerce and ERP platforms: order processing, customer management, payment-gateway integration.

Software Developer · Huawei Symantec Technology — Dec 2010 – Aug 2013

Led frontend and backend teams; multithreaded Linux C/C++; time-series anomaly detection in spectrum data (Kalman filter, HMM).

Software Developer · Topsec Network Security (002212.SZ) — Oct 2007 – Mar 2010

Linux intrusion detection with kernel-level packet analysis; DDoS detection.

§ 04 TECHNICAL SKILLS

MLOPS & CI/CD
GitHub Actions · Concourse · MLflow · Azure ML endpoints · Docker · Kubernetes · Automated testing & validation · Git

DATA & PIPELINES
Airflow · PySpark · AWS Glue · ETL · Snowflake · SQL · DynamoDB · Vector databases · Data warehousing

CLOUD
AWS (SageMaker, Lambda, Glue, S3, EC2) · Azure · GCP (Cloud Storage, Kubernetes)

ML & GENAI
PyTorch · TensorFlow / Keras · ONNX · Forecasting · LSTM · LLMs (LLaMA3, GPT) · VLMs · Claude Agent SDK · CLIP · Quantisation · Distillation · Computer vision

LANGUAGES
Python (10 yrs) · C/C++ (8 yrs) · C#/.NET · SQL

SERVICES & DELIVERY
Flask · REST / SOAP APIs · POS / payment integration · Alerting & ticketing · SAP / ERP integration · Agile / Scrum

§ 05 SELECTED RESEARCH

2026 · IN PREPARATION · POPETS
BodhiPromptShield: Pre-Inference Prompt Mediation for Suppressing Surface-Form Privacy Propagation in LLM Agent Pipelines.
github.com/mabo1215/BodhiPromptShield

2026 · UNDER REVIEW · ACM TACO
Shape Sets the Chunk: Selecting and Keying the Chunk of Chunked Linear-Recurrence GPU Kernels.
github.com/mabo1215/COREY_Transformer

2024 · MULTIMEDIA TOOLS AND APPLICATIONS
A privacy-preserving word embedding text classification model based on privacy boundary constructed by a deep belief network. 83(10): 30181–30206.

2021 · IEEE GLOBECOM
PPDTSA: Privacy-preserving deep transformation self-attention framework for object detection.

§ 06 EDUCATION

2019 – 2023 — **Doctor of Philosophy (PhD)**
Auckland University of Technology · New Zealand
Thesis: *Computational Methods in Machine Learning for Privacy Preservation*.

2010 – 2014 — **Doctoral research**
Northwestern Polytechnical University · China
Learning-augmented robust control for complex dynamic systems.

2003 – 2010 — **MSc, Computer Software & Theory · BEng, Data Science & E-Commerce**
Sichuan Normal University · China

§ 07 PATENTS · SERVICE

PATENTS Three granted / accepted patents (CNIPA): telephone-fraud prevention via voice semantic analysis; access-safety detection for virtualised users; multichannel network intrusion defence. Three software copyrights.

REVIEW Reviewer for AAAI, ACM TOMM, IEEE TCSVT, IEEE TETCI, IEEE Systems Journal, IEEE Access.